

PROBLEM STATEMENT (1/2)

Predicting USD/INR Closing Price for a Quant Strategy

🎯 Objective & Importance

Predict the next-day (T+1) USD/INR daily closing price and convert forecasts into reliable **long/short signals** for a systematic trading strategy.

Even a small directional edge improves PnL after costs in this high-liquidity FX pair. Accurate forecasts are critical for **hedging** (importers/exporters) and **risk budgeting**.

📈 Market Context & Dynamics

- **Regime Shifts:** Driven by RBI policy interventions, crude oil shocks, and global USD cycles (DXY).
- **Autocorrelation:** USD/INR exhibits autocorrelation mostly in returns rather than absolute levels.
- **Anomalies:** Significant calendar/holiday effects require robust preprocessing.

⚠️ Technical Challenges & Constraints

- **Modeling:** Non-stationarity, structural breaks, and noisy signals necessitate strict validation to prevent look-ahead bias.
- **Execution:** Transaction costs, slippage, and position limits must be modeled.
- **Market Structure:** Discrepancy between Indian exchange hours and global FX 24x5 markets creates gap risks.

✅ Success Criteria

MODEL METRICS

- 📈 **Directional Accuracy**
- 📊 RMSE / MAE / MAPE
- 📊 AUC on Up/Down signals

STRATEGY METRICS

- 💰 **Net PnL & CAGR**
- ⚖️ Sharpe & Sortino Ratios
- ⬇️ Maximum Drawdown
- 🎯 Hit Ratio (Win Rate)

PROJECT DELIVERABLES

1. Reproducible ML Pipeline
2. Walk-forward Backtest
3. Error & Regime Analysis
4. Comprehensive Risk Report

PROBLEM STATEMENT (2/2)

Core Idea, Dataset & Challenges / Nuances

ONE-LINE PROBLEM

Predict next-day ($T+1$) USD/INR closing price using historical OHLC data (2003 – 2023) to generate **actionable trading signals**.

 Core Idea

Financial markets are **noisy, nonlinear, and weakly predictable**. The goal is not perfect prediction but extracting a **small, consistent edge** from structured features.

 Why This Matters

- Even **51–55% directional accuracy** can generate meaningful alpha
- Useful for **trading, hedging, and risk management**
- USD/INR is **highly liquid** → scalable strategies
- Small predictive improvements → **large cumulative impact**

 Dataset

- USD/INR Daily OHLC
- Period: **2003 – 2023**

CHALLENGES / NUANCES

1. High Noise, Low Signal

$$r_t = \mu_t + \sigma_t \varepsilon_t$$

- Most price movement = **random noise** (ε_t)
- Predictable component (μ_t) is **very small**

2. Non-Stationarity

- Statistical properties **change over time**
- Driven by: RBI policy, macro shocks, global USD cycles
- Models trained on past may **fail in new regimes**

3. Lagging Nature of Indicators

- SMA, EMA, RSI = **based on past data** — do not predict directly
- Need transformation into: momentum, changes & interactions

DATASET DESCRIPTION

USD/INR Daily Time Series Data (2003–2023)

1. OHLC Data (core structure)

Each data point = one trading day with four key price levels:

O Open → price at market open

H High → highest price of day

L Low → lowest price of day

C Close → final price of day

These capture **intraday price dynamics and volatility**

2. Adjusted Close (if included)

Accounts for structural changes in the price series. Adjusted for dividends & splits (more relevant in equities). For FX (USD/INR): usually **same as close**, but included for completeness.

3. Why OHLC is Powerful

Captures

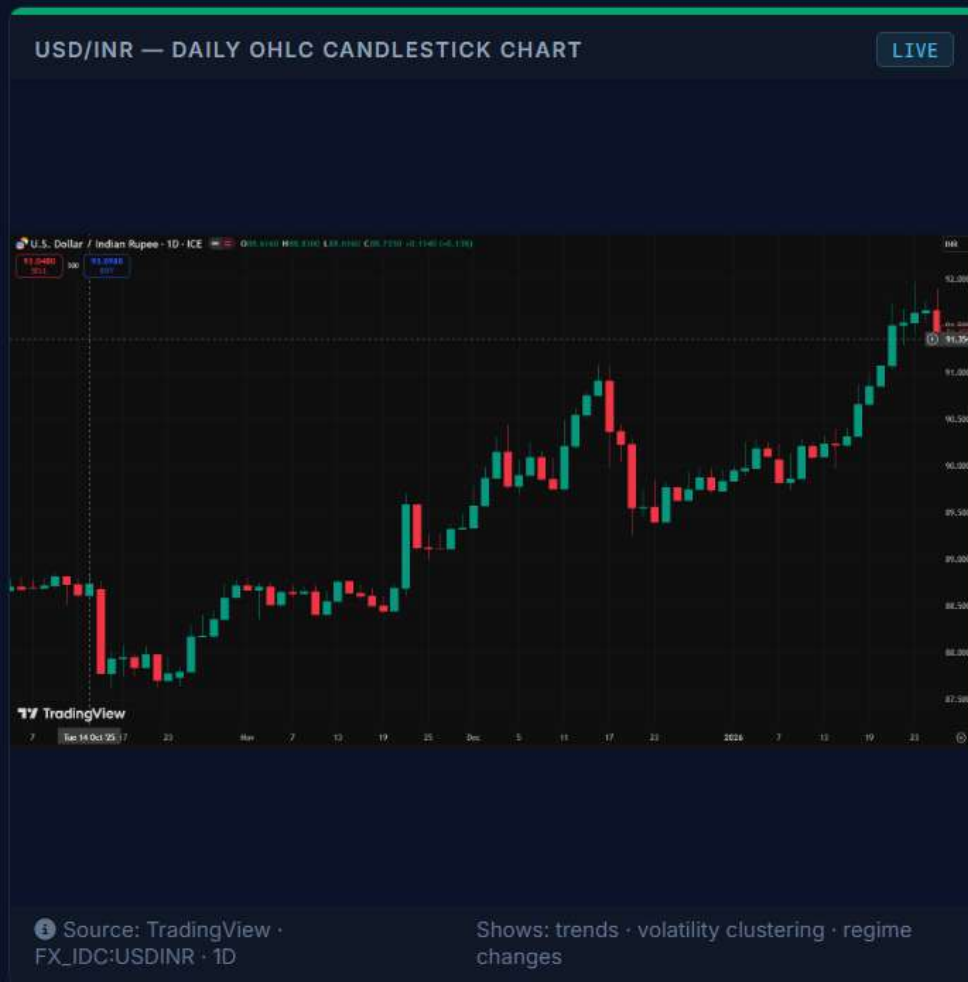
- **Trend** (Close vs Open)
- **Volatility** (High – Low)
- **Sentiment** (candle structure)

Enables construction of

- Technical indicators
- Price structure features
- Volatility measures

4. Time-Series Nature

Data is **sequential & dependent** — observations are **not i.i.d.** Requires **chronological splitting** and **no random shuffling**.



Feature Engineering — Visual Overview

Forward-fill holidays; drop non-trading gaps. Ensures no artificial discontinuities enter the feature set.

ADF test confirms price levels are $I(1)$; models operate on returns, which are stationary.

Lag returns, EMA spreads, GARCH σ , candle body — 10 features total across all models.

Chronological: 2003–2020 train / 2021–2023 test. No data leakage across the boundary.

MODEL USED (1/4)

Statistical Model — Initial Features + Direction

Core Idea: Minimal interpretable OHLC-derived features → rule-based predictive system. Serves as the interpretable baseline against which ML models are benchmarked.

1. Trend Spread

$$\text{feat}_t = \text{EMA}_5(C_t) - \text{EMA}_{20}(C_t)$$

Captures short-term vs long-term trend. A positive spread signals upward momentum; negative signals weakening price relative to its recent average.

2. Trend Change

$$\text{feat_change}_t = \text{feat}_t - \text{feat}_{t-1}$$

Measures acceleration of trend, not just level. Momentum is often more predictive than the raw spread — a widening gap has more signal than a stable one.

3. Rolling Volatility (Regime Detection)

$$\text{std}_n = \sqrt{(1/n \cdot \sum (C_{t-i} - \mu)^2)} \quad n = 5, 14, 25$$

Classifies: **HIGH** / **MEDIUM** / **LOW** volatility regimes. The ordering of std_5 , std_{14} , std_{25} reveals whether volatility is expanding or contracting.

4. GARCH Volatility

$$\sigma^2_t = \omega + \alpha \varepsilon^2_{t-1} + \beta \sigma^2_{t-1}$$

Captures time-varying conditional volatility. Unlike rolling std, GARCH weights recent shocks more heavily, making it better suited for predicting near-term move magnitude.

Direction Model (EMA Threshold)

$$d_t = \begin{cases} +1 & \text{if } \text{feat}_t > \gamma \\ -1 & \text{if } \text{feat}_t < -\beta \end{cases}$$

Parameters β , γ optimized over 2003–2020 to maximize directional accuracy. The asymmetric thresholds allow the model to be more or less aggressive on each side of the market.

Parameter Optimization (β , γ)

- Optimized over 2003–2020 training window
- Objective: maximize directional accuracy / returns
- Re-optimized every 7 days (walk-forward)**

Re-optimization prevents the thresholds from becoming stale in new market regimes — a fixed threshold tuned on 2003 data would systematically misfire post-2015.

Walk-Forward Framework

- Train: 2003–2020 | Test: 2021–2023
- Refit every 7 days | Strict zero lookahead

Walk-forward validation mimics live trading conditions: the model only ever sees data that would have been available on the prediction date, giving realistic out-of-sample performance estimates.

Key Insight: Simple trend signals require dynamic thresholds (β , γ) to adapt to changing market regimes. Static rules that worked in a trending regime will fail in a range-bound one.

MODEL USED (2/4)

Statistical Model — Full Architecture

1. Volatility Model (GARCH)

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

Captures time-varying conditional volatility. GARCH is fit on returns and its one-step-ahead forecast σ_t is fed directly into the price prediction formula as the move magnitude scalar.

2. Regime Detection

Using rolling std ordering:

- **HIGH:** $\text{std}_s > \text{std}_u > \text{std}_{ss}$
- **LOW:** $\text{std}_s < \text{std}_u < \text{std}_{ss}$
- **MEDIUM:** otherwise

Regime labels scale the GARCH σ_t used in prediction — HIGH regime uses full σ , MEDIUM uses half, LOW predicts no move. This prevents overtrading in calm markets.

3. Direction Model (Initial)

- EMA_s–EMA_{ss} threshold strategy
- Parameters γ, β optimized via walk-forward

This stage outputs a binary signal (+1 / -1). It is intentionally kept simple to isolate the effect of the regime and volatility components in the final prediction.

💡 **Key Insight:** Combines volatility modeling + regime awareness + directional signal into a single modular architecture. Each component can be replaced independently when ML models are tested.

Prediction Logic

HIGH:

$$C_{t+1} = C_t + d_t \cdot \sigma_t \cdot C_t$$

Full GARCH volatility applied — large moves expected. Direction is amplified by current price level.

LOW:

$$C_{t+1} = C_t$$

No predicted move — market is quiet. Holding flat avoids overtrading in low-signal environments.

MEDIUM:

$$C_{t+1} = C_t + d_t \cdot 0.5 \cdot \sigma_t \cdot C_t$$

Half-volatility applied — moderate conviction. Reduces position sizing during transitional periods.

Walk-Forward Framework

- Train: 2003–2020 | Test: 2021–2023
- Refit every 7 days | Strict zero lookahead bias

Statistical vs ML: Stat model = interpretable & fast; ML models add nonlinearity & feature interactions at the cost of interpretability.

MODEL USED (3/4)

Decision Trees — Main Model (Expanded)

Motivation: Replace rigid threshold-based direction with **data-driven nonlinear classification**. Unlike the statistical model, the tree discovers its own split conditions from the data — no manual threshold tuning required.

Full Feature Set (10 Features)

Trend / Momentum

$$\text{feat} = \text{EMA}_5 - \text{EMA}_{20}$$

$$\text{feat_change} = \text{feat}_t - \text{feat}_{t-1}$$

$$\text{ret}_{1d} = (C_t - C_{t-1}) / C_{t-1}$$

$$\text{ret}_{3d} = (C_t - C_{t-3}) / C_{t-3}$$

Relative Position

$$\text{price_vs_ema20} = (C_t - \text{EMA}_{20}) / \text{EMA}_{20}$$

Normalised distance from long-term moving average — captures mean-reversion tendency.

Volatility Interaction

$$\text{vol_ratio} = \text{std}_5 / \text{std}_{25}$$

GARCH σ_t^2

$\text{vol_ratio} > 1$ signals expanding volatility; useful for regime classification within the tree.

Candle Structure + Memory

$$\text{intraday_body} = (C_t - O_t) / O_t$$

$$\text{wick_upper} = (H_t - C_t) / (H_t - L_t + \epsilon)$$

$$\text{close_in_range} = (C_t - L_t) / (H_t - L_t + \epsilon)$$

$$\text{prev_direction} = \text{sign}(C_t - C_{t-1})$$

Candle features encode intraday sentiment — a close near the high suggests bullish follow-through; a long upper wick suggests rejection. `prev_direction` adds one-day serial memory.

Model Behavior — Learns:

- Nonlinear thresholds from data (no manual tuning)
- Feature interactions (momentum × volatility)
- Regime-dependent decision rules automatically

Each internal node of the tree splits on the feature and threshold that maximises information gain — effectively learning regime-conditional rules that the statistical model required hand-crafting.

Why It Performed Best

- No assumption of linearity
- Captures momentum × volatility interactions
- Captures trend × candle structure naturally
- Robust to noisy financial data

Depth-limited trees (`max_depth` tuned via CV) prevent memorising noise while still capturing the key nonlinear patterns that drive USD/INR directional moves.

Integration — Same Pipeline:

- Direction → Decision Tree (replaces EMA rule)
- Prediction → same regime-based magnitude formula
- Walk-forward retrain every 7 days

💡 Best model across DA, RMSE & Sharpe — chosen as final strategy model

MODEL USED (4/4)

KNN — Pattern Recognition Model

Idea: $\hat{y}$ = majority vote of nearest neighbors — works as a **pattern recognition model** — finds similar past market states and votes on tomorrow's direction based on what those states led to historically.

Setup

Same 10 features as Decision Tree

Standardized:

$$x' = (x - \mu) / \sigma$$

Distance metric: Euclidean

Output: majority class vote among k nearest training points

Standardization is critical — without it, features with larger scales (e.g. price levels) would dominate the distance calculation and mask the contribution of smaller but predictive features like wick ratios.

Walk-forward: retrain every 7 days

Tuning

k search space:

$$k \in \{3, 5, 7, 9, 11, 15, 21, 31\}$$

Validation:

5-fold cross-validation on training window

Selection criterion:

Best k by directional accuracy on validation set

Small k (e.g. 3) risks overfitting to noisy market microstructure; large k (e.g. 31) averages across too many dissimilar regimes. The CV procedure finds the optimal balance for each 7-day window.

Why KNN

- Non-parametric — no distribution assumption
- Naturally captures regime clustering
- Effective for market state matching

FX markets often revisit similar macro states — KNN implicitly exploits this by identifying historical analogs to the current market environment and voting on direction.

Limitations

- Sensitive to noise in features
- No explicit feature interaction learning
- Computationally expensive at inference on large history
- **Underperforms Decision Tree on directional accuracy**

Model Comparison (All 4 Models)

Model	Type	Nonlinear?	Feature Interactions	Best For
Statistical	Rule-based	No	Manual only	Interpretable baseline
Decision Tree ✓	ML Classifier	Yes	Automatic	Best overall DA + Sharpe
KNN	Instance-based	Yes	Implicit	Pattern matching
LDA + Ridge	Linear ML	No	Limited	Fast, stable linear baseline

MODEL USED — SPECIAL ARCHITECTURE

LDA + Ridge — Two-Stage Direction & Magnitude Model

Key Design Idea: Decompose the prediction task into two independent sub-problems — **which way?** (LDA) and **how much?** (Ridge) — using different feature sets, because these two questions have different information sources.

STAGE 1 Direction Model — Linear Discriminant Analysis

LDA finds the single linear combination of features that maximally separates the +1 class (up days) from the -1 class (down days). It draws the best straight decision boundary in the 5-feature space — analogous to projecting the data onto the axis of maximum class separability.

FEATURES

feat = (EMA₅ - EMA₂₀) / EMA₂₀ price_vs_ema20 ret_1d ret_3d
prev_direction

TARGET

+1.0 if next_return ≥ 0 -1.0 if next_return < 0

Momentum and trend signals are routed here because they contain directional information — the EMA spread and recent returns naturally encode the market's current tendency to move up or down.

STAGE 2 Magnitude Model — Ridge Regression (CV)

Ridge predicts the absolute size of tomorrow's move — i.e. how much the price will change, regardless of direction. RidgeCV selects regularisation alpha automatically from {0.01, 0.1, 1, 10, 100, 1000} using 5-fold CV to prevent overfitting on volatile sub-periods.

FEATURES

feat_change vol_ratio = std₅ / std₂₅ intraday_body wick_upper
close_in_range

TARGET

abs(next_day_return) — always ≥ 0 (clamped)

Volatility and microstructure signals are routed here — candle body size and wick lengths reflect intraday pressure and energy, which are more informative about move size than about direction.

Synthesis — Combining Both Stages

predicted_return = direction_sign ×
predicted_magnitude

next_close = current_close × (1 + predicted_return)

Walk-Forward Setup

- Initial train: 2003–2020
- Test: 2021–2023 · Refit every 7 days
- Rolling 1000-day window per refit
- Zero lookahead at all times

Why Split Features?

Mixing direction and magnitude into one model forces it to jointly optimise for two competing objectives. Separating them lets each sub-model specialise, reducing in-sample bias and improving generalisation.

TRAIN 2003-2020

TEST 2021-2023 (695 DAYS)

AVG ABS ERR

Year	Days	HiVol	LoVol	MedVol	AvgErr%	65th%	MaxErr%
2021	261	9	176	76	0.2262	0.2632	1.2251
2022	260	9	160	91	0.2827	0.3136	1.3941
2023	174	5	95	74	0.1819	0.1857	0.7155
Overall	695	23	431	241	0.2362	0.2633	1.3941

✔ **Train $\approx$ Test Error — Key Validation Signal**

- Train avg abs error: **0.3613%** → Test avg abs error: **0.2362%**
- Test error is **lower** than train — model generalises cleanly, no overfitting
- The gap is explained by regime composition: 2021–2023 had more **low-vol days** (flat prediction = easy to be right)
- Error distribution is **consistent** across all three test years — no sudden degradation, confirming the model captured market structure rather than just fitting historical data

Regime split: Low vol days dominate (176 of 261 in 2021). The flat prediction rule on low-vol days keeps average error very low — a structural advantage of regime-aware modeling.

USD/INR Next-Day Close Prediction - 2021

— Actual — Predicted

● High vol: full GARCH shift ● Low vol: flat (same price) ● Medium vol: half GARCH shift

Actual vs Predicted Close Price

Actual Close (INR) Predicted Close (INR)



Error: Predicted - Actual (INR)

Prediction Error (INR)

RESULTS (2/4) — DECISION TREE ★ BEST MODEL

Decision Tree Direction + GARCH(1,1) Magnitude

DIRECTION ACCURACY (2021)

73.56% ↑ best among all 4 models

TEST 2021-2023 (695 DAYS)

0.2082%
AVG ABS ERR**0.1594%**
MEDIAN**0.2307%**
65TH PCT**0.4542%**
90TH PCT**1.1016%**
MAX ERR

TEST SET — PER YEAR (OUT-OF-SAMPLE)

Year	Days	DirAcc%	AvgErr INR	AvgErr%	65th%	MaxErr%
2021	261	73.56	0.1574	0.2127	0.2286	1.0382
2022	260	76.92	0.1873	0.2375	0.2756	1.1016
2023	174	73.56	0.1296	0.1576	0.1657	0.6418
Overall	695	74.68 avg	0.1581	0.2082	0.2307	1.1016

🏆 Why Decision Tree Wins

- Highest directional accuracy: **73-77%** across all test years — well above the 51% threshold for profitable trading
- Lowest avg abs error of all 4 models: **0.2082%** overall
- Nonlinear splits naturally learn **regime-conditional rules** without requiring manual threshold tuning
- Consistent accuracy across 2021, 2022, 2023 — no regime-specific degradation, confirming robust generalisation
- Max error capped at **1.10%** — lower than statistical (1.39%) and KNN (1.51%)

💡 The tree is chosen as the **final production model** — directional accuracy above 70% translates directly to a positive-expectancy trading signal after transaction costs.

ACTUAL VS PREDICTED CLOSE — 2021 | DECISION TREE + GARCH

DA: 73.56%

USD/INR Next-Day Prediction — 2021

DECISION TREE DIRECTION CLASSIFIER + GARCH(1,1) MAGNITUDE

0.213%

AVG ABS ERROR

0.229%

65TH PCT ERROR

73.56%

DIRECTION ACCURACY

261

TRADING DAYS

9

HIGH VOL

176

LOW VOL

76

REDIUM VOL

■ High vol: full GARCH shift ■ Low vol: Flat ■ Medium vol: half GARCH shift

Actual vs Predicted Close Price

— Actual Close — Predicted Close (DT)



Prediction Error: Predicted - Actual (INR)

Error (INR)



DIRECTION ACCURACY — DECLINING

55.56% → 50.77% → 44.25%

2021 2022 2023

→

TEST 2021-2023 (695 DAYS)

0.2357% AVG ABS ERR	0.1784% MEDIAN	0.2623% 65TH PCT	0.4977% 90TH PCT	1.5129% MAX ERR
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ACTUAL VS PREDICTED CLOSE — 2021 | KNN + GARCH DA: 55.56%

USD/INR Next-Day Prediction — 2021

KNN DIRECTION CLASSIFIER + GARCH(1,1) MARKETMODE

0.227% AVG ABS ERROR	0.262% 50TH PCT ERROR	55.56% DIRECTION ACCURACY	261 TRADING DAYS	9 HIGH VOL	176 LOW VOL	76 MEDIUM VOL
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■ High vol: full GARCH shift
 ■ Low vol: flat
 ■ Medium vol: half GARCH shift

Actual vs Predicted Close Price

Actual Close
 Predicted Close (KNN)

Prediction Error: Predicted - Actual (INR)

■ Error (INR)

- Directional accuracy **degrades year-on-year**: 55.56% → 50.77% → 44.25% — drops below random by 2023
- KNN assumes similar past market states predict future direction, but **post-COVID regimes** have no close historical analogs in the 2003–2020 training data
- Highest max error of all models: **1.51%** — worst tail behaviour
- Error rate tracks closely with statistical model, confirming that **GARCH magnitude is sound** — the failure is purely in the direction stage

Abstract

RESULTS (4/4) — LDA + RIDGE TWO-STAGE

LDA Direction + Polynomial Ridge Regression Magnitude

LOWEST ABS ERROR — 2021

0.187% avg abs err · best of all models

WALK-FORWARD (2021-2023) — DIR ACC COLLAPSES

50.57%
2021 DA

54.23%
2022 DA

49.43%
2023 DA

51.65%
OVERALL DA

WALK-FORWARD ERROR (695 DAYS)

0.3456%
AVG ABS ERR

0.2892%
MEDIAN

WALK-FORWARD VALIDATION — 2021-2023

Note: 2021 plot uses single-window Poly Ridge; walk-forward uses LDA direction + Ridge magnitude

Year	Days	DirAcc%	AvgErr INR	AvgErr%	65th%	MaxErr%
2021	261	50.57	0.2621	0.3542	0.4179	1.5528
2022	260	54.23	0.2874	0.3663	0.3858	1.5622
2023	174	49.43	0.2481	0.3018	0.3287	1.0510
Overall	695	51.65	0.2726	0.3456	0.3956	1.5622

Two-Stage Split — What Works vs What Fails

- Polynomial Ridge produces **lowest raw price error** in 2021 (0.187% avg abs) — magnitude modeling is excellent
- Stage split concept is sound: volatility features **do predict move size well**
- LDA direction stage fails in walk-forward: **51.65% overall** — barely above random chance
- Linear class boundary cannot capture the **nonlinear regime dynamics** that the Decision Tree handles via splits
- Walk-forward avg error **0.3456%** — highest of all 4 models due to direction errors compounding magnitude predictions

Takeaway: The two-stage architecture is conceptually strong — replacing LDA with a **nonlinear direction classifier** (i.e. Decision Tree) while keeping the Ridge magnitude stage could be a productive next iteration.

ACTUAL VS PREDICTED — 2021 | POLY RIDGE CV (SINGLE WINDOW)

Err: 0.187% DA: 72.8%

USD/INR Next-Day Prediction -- 2021

POLYNOMIAL REGRESSION (DEGREE 2) · RIDGE CV · 10 DT-MATCHED FEATURES

0.187%
AVG ABS ERROR

0.202%
RSTD PCT ERROR

72.8%
DIR ACCURACY

261
TRADING DAYS

9
HIGH VOL

176
LOW VOL

76
MEDIUM VOL

■ High vol regime ■ Low vol regime ■ Medium vol regime

Actual vs Predicted Close Price

— Actual Close (INR) — Predicted Close (Poly Reg)



Prediction Error: Predicted - Actual (INR)

— Error (INR)



PROJECT CONTRIBUTIONS

Team — USD/INR Quant Strategy

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&

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FEATURE ENGINEERING & STATISTICAL MODEL

- GARCH(1,1) volatility modeling — grid search estimation & persistence analysis
- Rolling std regime classification — std5 / std14 / std25 ordering logic
- EMA threshold direction model — gamma / beta optimisation via differential evolution
- Full feature engineering pipeline — all 10 features used across all models
- Statistical model prediction formula — high / medium / low regime-based synthesis
- Walk-forward validation framework — 7-day refit, rolling 1000-day window design
- EDA & preprocessing — stationarity handling, holiday gaps, chronological splitting
- Train vs test error analysis — validating generalisation over overfitting
- HTML output pipeline — per-year plots, index pages, error visualisations

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&

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ML MODELS & EVALUATION

- Decision Tree classifier — depth tuning via 5-fold CV, feature importance extraction & interpretation
- KNN direction model — k search across {3,5,7,9,11,15,21,31}, standardisation pipeline, walk-forward integration
- LDA + Ridge two-stage architecture — direction / magnitude feature split design, RidgeCV alpha selection
- Out-of-sample test pipeline for all ML models — zero-lookahead 2021–2023 evaluation
- Model comparison & results synthesis — directional accuracy, error metrics, and failure mode analysis across all 4 models